

MATH-1014 T10A

Tutorial 8: Approximate Integration & Polar Coordinates

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Today's Roadmap

1 Approximate Integration

2 Polar Coordinates & Curves

Review: Quadrature Rules

We use numerical integration when finding an antiderivative is impossible (e.g., $\int e^{x^2} dx$) or when we only have discrete data points.

Core Formulas ($\Delta x = \frac{b-a}{n}$)

- **Midpoint Rule:** $M_n = \Delta x \sum_{i=1}^n f(\bar{x}_i)$, where $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i)$.
- **Trapezoidal Rule:** $T_n = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]$.
- **Simpson's Rule (n is even):** Uses parabolic interpolation.
 $S_n = \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{n-1}) + f(x_n)]$.

Review: Error Bounds

To guarantee accuracy, we use error bounds derived from higher-order derivatives. Let $|f''(x)| \leq k$ and $|f^{(4)}(x)| \leq K$ for $a \leq x \leq b$:

Error Bound Formulas

$$|E_T| \leq \frac{k(b-a)^3}{12n^2}, \quad |E_M| \leq \frac{k(b-a)^3}{24n^2}$$
$$|E_S| \leq \frac{K(b-a)^5}{180n^4}$$

Remark: The Midpoint error is roughly half the Trapezoidal error. Simpson's Rule converges much faster ($\mathcal{O}(h^4)$) due to higher-order approximations.

Problem 1: Error Bounds & Optimization

Problem 1. Approximate the following integral using **Simpson's Rule**:

$$I = \int_0^1 e^{-x^2} dx$$

Find the minimum number of subintervals n required to guarantee an absolute error of less than 10^{-4} .

Hint

You need to bound the 4th derivative $|f^{(4)}(x)| \leq K$ on $[0, 1]$. Remember that n must be an even integer for Simpson's Rule.

Solution 1 (Part 1/2): Finding K

Let $f(x) = e^{-x^2}$. Computing successive derivatives yields:

$$f'(x) = -2xe^{-x^2}$$

$$f''(x) = (4x^2 - 2)e^{-x^2}$$

$$f'''(x) = (12x - 8x^3)e^{-x^2}$$

$$f^{(4)}(x) = (16x^4 - 48x^2 + 12)e^{-x^2}$$

To maximize $|f^{(4)}(x)|$ on $[0, 1]$, let $g(x) = 16x^4 - 48x^2 + 12$. Since $e^{-x^2} \leq 1$ on $[0, 1]$, we analyze $|g(x)|$. $g'(x) = 32x(2x^2 - 3)$, so the only critical point in $[0, 1]$ is $x = 0$.

- At $x = 0$: $|f^{(4)}(0)| = 12 \times 1 = 12$.
- At $x = 1$: $|f^{(4)}(1)| = |-20|e^{-1} \approx 7.36$.

Thus, we can safely take the maximum bound $K = 12$.

Solution 1 (Part 2/2): Solving for n

Using the error bound formula for Simpson's Rule:

$$|E_S| \leq \frac{K(b-a)^5}{180n^4} < 10^{-4}$$

Substitute $K = 12$, $a = 0$, and $b = 1$:

$$\frac{12(1-0)^5}{180n^4} < 10^{-4} \implies \frac{1}{15n^4} < 10^{-4}$$

Solving for n :

$$n^4 > \frac{10000}{15} \approx 666.67 \implies n > \sqrt[4]{666.67} \approx 5.08$$

Since n must be an **even** integer for Simpson's Rule, the minimum required number of subintervals is:

$$\boxed{n = 6}$$

Review: Polar Coordinates & Symmetry

Cartesian to Polar Connection

For a point $P(r, \theta)$, $x = r \cos \theta$ and $y = r \sin \theta$.

Symmetry Tests

- **Polar Axis (x -axis):** Unchanged when $\theta \rightarrow -\theta$.
- **The Pole (Origin):** Unchanged when $r \rightarrow -r$ or $\theta \rightarrow \theta + \pi$.
- **Vertical Line ($\theta = \pi/2$):** Unchanged when $\theta \rightarrow \pi - \theta$.

Review: Tangents to Polar Curves

For a polar curve $r = f(\theta)$, $x = f(\theta) \cos \theta$ and $y = f(\theta) \sin \theta$. Using the chain rule, the slope of the tangent line is:

Derivative Formula

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

Critical Note: A horizontal tangent occurs when $\frac{dy}{d\theta} = 0$ provided $\frac{dx}{d\theta} \neq 0$. If $\frac{dy}{d\theta} = \frac{dx}{d\theta} = 0$ at the same point, you **must** evaluate $\lim \frac{dy}{dx}$.

Problem 2: Tangent Singularities

Problem 2. Consider the cardioid given by the polar equation:

$$r = 1 + \cos \theta, \quad 0 \leq \theta \leq 2\pi$$

Find all points (r, θ) on the curve where the tangent line is strictly **horizontal**.

Caution

You will encounter a singular pole at $\theta = \pi$ where the curve forms a cusp. Evaluate $\lim_{\theta \rightarrow \pi} \frac{dy}{dx}$ using L'Hôpital's rule to determine the true geometric behavior.

Solution 2 (Part 1/2): Finding Critical Points

Express y and x in terms of θ to find their derivatives:

$$y = r \sin \theta = (1 + \cos \theta) \sin \theta$$

$$\frac{dy}{d\theta} = -\sin^2 \theta + \cos \theta(1 + \cos \theta) = 2 \cos^2 \theta + \cos \theta - 1$$

Setting $\frac{dy}{d\theta} = (2 \cos \theta - 1)(\cos \theta + 1) = 0$ yields $\theta \in \left\{ \frac{\pi}{3}, \pi, \frac{5\pi}{3} \right\}$.

Next, we check the denominator $\frac{dx}{d\theta}$:

$$x = r \cos \theta = \cos \theta + \cos^2 \theta$$

$$\frac{dx}{d\theta} = -\sin \theta - 2 \cos \theta \sin \theta = -\sin \theta(1 + 2 \cos \theta)$$

Solution 2 (Part 2/2): Limit Analysis

At $\theta \in \{\frac{\pi}{3}, \frac{5\pi}{3}\}$, $\frac{dx}{d\theta} \neq 0$. Thus, horizontal tangents occur at:

$$(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{3}\right) \text{ and } \left(\frac{3}{2}, \frac{5\pi}{3}\right)$$

At the cusp $\theta = \pi$, both $\frac{dy}{d\theta} = 0$ and $\frac{dx}{d\theta} = 0$. We evaluate the limit:

$$\lim_{\theta \rightarrow \pi} \frac{dy}{dx} = \lim_{\theta \rightarrow \pi} \frac{(2 \cos \theta - 1)(\cos \theta + 1)}{-\sin \theta(1 + 2 \cos \theta)}$$

Since $(2 \cos \theta - 1) \rightarrow -3$ and $(1 + 2 \cos \theta) \rightarrow -1$, this simplifies to:

$$3 \lim_{\theta \rightarrow \pi} \frac{\cos \theta + 1}{-\sin \theta} \stackrel{\text{L'H}}{=} 3 \lim_{\theta \rightarrow \pi} \frac{-\sin \theta}{-\cos \theta} = 3 \left(\frac{0}{1} \right) = 0$$

The limit is 0, confirming a horizontal tangent at $\theta = \pi$.

Final Answer: $(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{3}\right), \left(\frac{3}{2}, \frac{5\pi}{3}\right), (0, \pi)$.

Review: Area and Arc Length in Polar Coordinates

Area of a Polar Region

The area A bounded by the curve $r = f(\theta)$ and rays $\theta = a, \theta = b$ is:

$$A = \int_a^b \frac{1}{2} r^2 d\theta = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta$$

Arc Length of a Polar Curve

The length L of $r = f(\theta)$ for $a \leq \theta \leq b$ is:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Problem 3: Area Between Polar Curves

Problem 3. Find the exact area of the region lying **inside** the circle $r = 3 \sin \theta$ and **outside** the cardioid $r = 1 + \sin \theta$.

Hint

1. Find the intersection points to establish the limits of integration.
2. Leverage symmetry (calculate the area in Q1 and multiply by 2).
3. Use the area formula $A = \frac{1}{2} \int (r_{outer}^2 - r_{inner}^2) d\theta$.

Solution 3 (Part 1/2): Finding Intersections

Equate the two equations to find the intersection points:

$$3 \sin \theta = 1 + \sin \theta \implies 2 \sin \theta = 1 \implies \sin \theta = \frac{1}{2}$$

For $\theta \in [0, \pi]$, the intersections are at $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$.

Since both curves are symmetric with respect to the y -axis ($\theta = \pi/2$), we can integrate from $\frac{\pi}{6}$ to $\frac{\pi}{2}$ and double the result:

$$A = 2 \int_{\pi/6}^{\pi/2} \frac{1}{2} [(3 \sin \theta)^2 - (1 + \sin \theta)^2] d\theta$$

Solution 3 (Part 2/2): Evaluating the Integral

Expand the integrand and apply $8 \sin^2 \theta = 4(1 - \cos 2\theta)$:

$$\begin{aligned}(3 \sin \theta)^2 - (1 + \sin \theta)^2 &= 8 \sin^2 \theta - 2 \sin \theta - 1 \\ &= (4 - 4 \cos 2\theta) - 2 \sin \theta - 1 \\ &= 3 - 4 \cos 2\theta - 2 \sin \theta\end{aligned}$$

Now, evaluate the integral directly:

$$\begin{aligned}A &= \int_{\pi/6}^{\pi/2} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\pi/6}^{\pi/2}\end{aligned}$$

Substituting the limits ($\pi/2$ and $\pi/6$):

$$A = \left(\frac{3\pi}{2} - 0 + 0 \right) - \left(\frac{\pi}{2} - \sqrt{3} + \sqrt{3} \right) = \pi$$

Final Area: $A = \boxed{\pi}$

THANK YOU!

